Al-Balqa Applied University

جامعة البلقاء التطبيقية

Midterm Exam

Course Title	General Chemistry lab		Course Code	30206111	
Semester	2nd		Academic Year	2024-2025	
Exam Date	6/4/2025	Day	Sunday	Time	2:30-3:20
Student Name	[redacted]		Student ID No.	Section	monday 8:30 - 11:30

اسم المدرس: Dr. Heba

GENERAL INSTRUCTIONS

1. Students are not allowed to leave the exam room during the first 30 minutes of the exam.
2. Late students are not permitted to complete their exams after the first 30 minutes unless authorized by the proctor (no students left the exam session), knowing that no additional time will be granted.
3. Students must use a blue or black pen. Pencils are not allowed.
4. Consumption of any food or beverage is prohibited except in medical cases.
5. All electronic devices (e.g., cell phones) are prohibited.
6. All cell phones must be turned off and placed on the invigilator's desk in the exam room.
7. All notes and reading materials must be placed on the invigilator's desk in the exam room.
8. Students are not permitted to speak to or ask questions of their classmates. If a student has a question, he or she must raise his or her hand to be assisted by the proctor.
9. Write the full information required on this page before commencing your exam.
10. Students are not allowed to leave the room once the exam commences for any reason unless it is absolutely necessary.
11. Reported student cheating cases, misbehavior, or other student misconduct on examination sessions will be reported to the Investigations Committee as per internal bylaws.

I have understood and abide with the exam policy. **Student's signature:**

Question No.	Question Type	CILOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
1 – 8	Multiple Choice	CLIO	53.3 %	16	16	28
9 - 15	Assay	CLIO	46.7 %	14	12	
Total			100 %	30		
Moderation Review						
Comments						
Marks Reviewed by					Signature & Date	

Part 1

Question	1	2	3	4	5	6	7	8
Answer	D	C	A	D	B	C	C	D

Part 2

Q	Answer	Q	Answer	Q	Answer
1	funnel	2	A X	3	color change
4	fume hood	5	it will increase	6	17.5 moles
7	56.93%				

PART1

Choose the most correct answer for the following questions and transfer it in CAPITAL letters to table on the cover page.

Q1-What should you do if you accidentally spill a chemical on yourself?

A. Wait until the end of the experiment to clean it up
B. Ignore it, as small spills are not harmful
C. Ask someone else to clean it up for you
D. Clean it up immediately using the appropriate procedure and notify your supervisor

Q2-Which of the following is true about eating or drinking in the lab?

A. It is allowed as long as you are careful not to spill
B. It is allowed during non-experimental periods
C. It is strictly forbidden at all times
D. It is allowed in designated areas of the lab

Q3-When using a Bunsen burner, which statement is correct?

A. Always point the open end of the tube away from yourself and others
B. Place the Bunsen burner near flammable materials for easy access
C. It's not necessary to have a fire extinguisher nearby while using the burner
D. It's okay to leave the Bunsen burner unattended while it's lit

Q4 What is the mass of a cubic block (cube) of silicon of side 9.654 cm length, given that density of silicon (Si) is 2.336 g/cm³?

A. 1051.2g B.22.55g C.385.3g D.2102.4g

Q5-Which one of the following salts is water insoluble?

A. H_2SO_4 B. $CaCO_3$ C. K_3PO_4 D. NaCl

Q6-What is the function of a capillary tube in a boiling point determination?

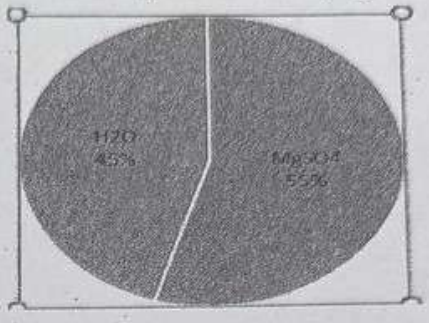
A. To increase the vapor pressure of the liquid.
B. To condense the vapor.
C. To provide a nucleation site where you can easily observe the bubble formation.
D. To measure the volume of vapor.

Q7-A student measures the boiling point of liquid water to be 100°C at an atmospheric pressure of 1.0 atm. What can be concluded?

A. The liquid water contains soluble impurities.
B. The liquid water contains insoluble impurities.
C. The liquid water is pure.
D. A&B

Q8-Calculate water of crystallization (X f)or the salt which shown beside knowing that M.m of ($MgSO_4$ =120.3g/mol and H_2O= 18g/mol)

A. 7
B. 8
C. 9
D. 5



Answer the following questions and *transfer your answer to the table on the cover page*

Q		Answer
1	**What is the name of the glassware shown in the picture?** 	funnel
2	**Two liquids, A and B, have densities 1.75 grams per milliliter and 1.14 grams per milliliter, respectively. When both liquids are poured into a container, one liquid floats on top of the other. Which liquid is on top?**	A
3	**What physical observation do you see when I_2 is mixed with benzene C_6H_6?**	color change
4	**What safety equipment is usually needed for experiments generating poisonous and harmful gases?**	fume hood
5	**During experiment of water of crystallization, if some of the anhydrous salt spilled out of the crucible before being weighed, what potential error might occur on the value of X?**	it will increase
6	**If a hydrated salt has the formula CuSO4·5H2O, how many moles of water are present per 3.5 mole of CuSO4?**	17.5 moles
7	**A hydrate of magnesium chloride is heated, and the following data are collected:** **mass of crucible = 22.130 grams** **mass of crucible + hydrate = 25.290 grams** **mass of crucible and contents after heating = 23.491 grams** **What is the % of water in the sample ?**	56.93%

Good LUCK